

Zebrafish anterior hindbrain / dIPN: a vertebrate ring attractor for heading direction

Abstract

We study a vertebrate ring attractor for heading direction at single-neuron anatomical resolution: the GABAergic $r1\pi$ population of the larval-zebrafish anterior hindbrain (aHB), whose axons arborise in the dorsal interpeduncular nucleus (dIPN). Petrucco et al. [2023] demonstrate a sinusoidal activity bump that rotates with the fish’s swims and is otherwise stable over many seconds, with reciprocal inhibitory connectivity in dIPN providing the long-range ring-shaping inhibition. We assemble a connectome-constrained training pipeline for this dataset: skeletons, ROI meshes and the (post, pre) adjacency matrix between the 731 HD-circuit neurons are fetched from the *neuprint-fish2* server (dataset *fish2*), and the sign-locked recurrent and message-passing GNN parameterisations of a swim-integration task are trained under an all-inhibitory Dale convention for the bump population. This stub document collects the anatomy and connectome summary; the modelling sections will be filled in once the zebrafish task generator and training runs are in place.

1 Introduction

In the larval zebrafish, head-direction information is carried by a population of ~ 50 – 100 GABAergic neurons (the $r1\pi$ cells) located in rhombomere 1 of the anterior hindbrain [Petrucco et al., 2023]. A sinusoidal bump of activity travels around this network in a direction opposite to the fish’s swims, persists for tens of seconds in the absence of motor activity, and tracks an estimated heading angle whose magnitude per swim ($\approx \pi/4$ rad) matches the behaviourally measured turn. The connectivity scheme that supports the bump — short-range excitation and long-range, out-of-phase inhibition — is the classical ring-attractor architecture, and electron-microscopy reconstructions of the $r1\pi$ axons show reciprocal inhibitory connectivity in the dIPN consistent with this picture.

Cell-type vocabulary. Tab. 1 lists the four functional categories of HD-circuit neurons exposed by the *fish2* neuprint server, together with their roles and *fish2* type prefixes.

Role	Zebrafish population	<i>fish2</i> type prefix	<i>n</i>
Bump (HD ring)	dorsal IPN ($r1\pi$)	<i>IPNd*</i>	338
Bump (HD ring, subset)	dorsal-subset IPN	<i>IPNds*</i>	105
Long-range inhibition	intrinsic to dIPN	<i>IPNd*</i> (Dale: inh.)	—
Velocity afferents	habenula $\rightarrow$ IPN	<i>RIPN*</i>	200
Visual / context afferents	pretectum $\rightarrow$ IPN	<i>pt-IPN*</i>	88
<i>total HD-circuit neurons</i>			731

Table 1: HD-circuit cell-type families identified in Petrucco et al. [2023] and used throughout this document. The HD ring is all-inhibitory: the bump cells themselves provide the long-range inhibition, hence the dash in the count column for that row. The afferent split (*RIPN* / *pt-IPN*) carries the swim-triggered and visual drives respectively; *RIPN** is functionally implicated as a swim-triggered driver of the bump, *pt-IPN** carries visual context. The asterisk in *IPNd** etc. is a wildcard: each row groups the many fine-grained *fish2* subtypes whose name starts with that prefix (e.g. *IPNd13B*, *IPNd17A*, *IPNdp01*, ... — 31 subtypes in total).

2 Anatomy and connectome

Data. Skeletons, ROI meshes and synaptic connections are fetched from *neuprint-fish2.janelia.org*, dataset *fish2. figures/zebrafish/fetch_zebrafish_anatomy_HD.py* writes SWC skeletons and OBJ meshes to *zebrafish_anatomy_HD/*; *fetch_zebrafish_connectivity_HD.py* writes *neurons.csv* (731 rows: 443 *IPNd** + *IPNds** ring cells, 200 *RIPN**, 88 *pt-IPN**) and *connections.csv* (14391 directed edges) to *zebrafish_connectome_HD/*. The loader *load_zebrafish_hd_connectome()* in *src/connectome_gnn/generators/constr_data.py* reads the two CSV files and returns a dictionary with the canonical keys *N*, *J_effective*, *neuron_types*, *type_names*, *n_ring*, *ring_ix*, *afferent_subpop_ix*. The *ZebrafishHdTaskRNN* class (registered as *zebrafish_hd_si* in *src/connectome_gnn/models/zebrafish*) implements the sign-locked dynamics, the 4-scalar afferent gate, the HD-ring-only readout, and the regulariser machinery described in Sect. 3.

The loader fields are:

- *n_ring* = 443: number of dorsal-IPN HD cells (*IPNd** + *IPNds**), i.e. the $r1\pi$ ring.
- *ring_ix*: ring ordering of the HD cells, computed from the mediolateral soma coordinate *somaLocationX* (Petrucco’s Fig. 1g right panel shows the $r1\pi$ cells lie on an anatomical ring along this axis); cells are binned into 16 ring positions for the readout decoder.
- *afferent_subpop_ix* = {*RIPN_L*, *RIPN_R*, *ptIPN_L*, *ptIPN_R*} split by side from the *instance* suffix; sizes {102, 101, 41, 44}.

Dale convention. The $r1\pi$ cells (*IPNd** and *IPNds**) are GABAergic [Petrucco et al., 2023]; their outgoing weights are sign-flipped and amplified by $5\times$. *RIPN** and *pt-IPN** have no Dale annotation on the *fish2* server and are kept positive (excitatory). The resulting signed matrix is rescaled so $\max(\text{Re}(\lambda)) = 0.9$, a spectral-radius convention that keeps the linearised recurrent operator inside the stability margin used throughout the training runs.

3-D anatomy. Fig. 1 shows the four HD-circuit cell-type families (*IPNd**, *IPNds**, *RIPN**, *pt-IPN**) rendered in 3-D against the Habenula / dIPN / vIPN ROI meshes. The *IPNd*/*IPNds* arbors

form the compact midline ring-attractor structure; *RIPN* fibres descend from the habenula and *pt-IPN* fibres sweep in from the lateral pretectum.

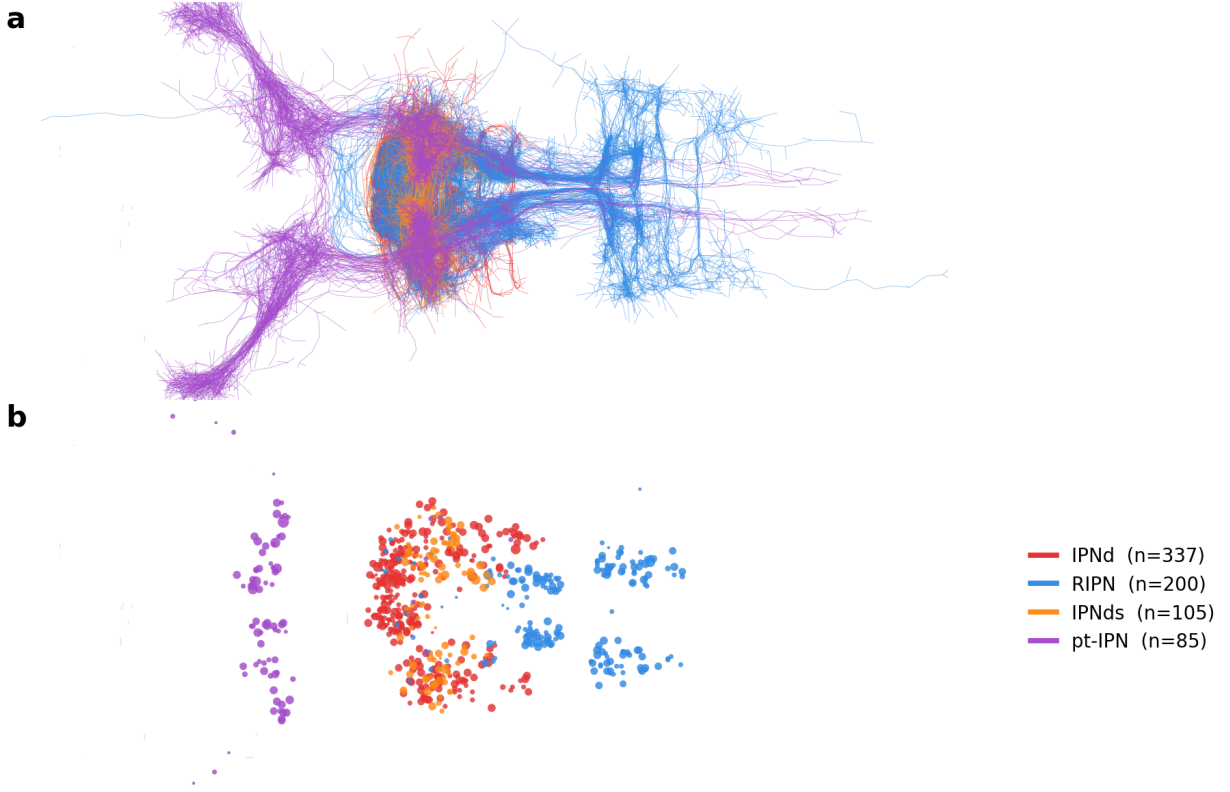


Figure 1: **Zebrafish HD-circuit anatomy.** (a) 727 skeletons from *neuprint-fish2* grouped into the four functional categories of Tab. 1: *IPNd** (red, dIPN HD ring), *IPNds** (orange, dorsal-subset), *RIPN** (blue, habenula → IPN afferents), *pt-IPN** (purple, pretectum → IPN afferents). Dorsal view, elevation +90°, azimuth -0.5°. ROI meshes: *Habenula_dIPN*, *Habenula_vIPN*, *dIPN(L1)/(R1)*, *dsIPN(L)/(R)*, *iIPN(L/R){1,2,3}*, *Pretectum*, *Habenula* (faint silhouette outlines). (b) Same view restricted to the cell bodies, with each soma drawn from the SWC soma node (the only node whose radius is $\gg$ the segmental baseline). The two panels show *spatially disjoint* compartments of the same neurons: the dense central region of (a) is the dIPN / iIPN *neuropil* — a region of axo-dendritic arborisation containing no cell bodies — and the somata in (b) sit in the cortex around it, each connected to its neuropil arbor by a thin primary-neurite tract visible in (a). The HD-ring substrate is therefore not a ring of cell bodies but the neuropil into which IPNd / RIPN / pt-IPN arbors converge.

Connectome summary. Fig. 2 summarises the connectome as the signed W^{con} (panel a, z -scored over non-zero entries and clipped to ± 3) and its binary support $\text{supp}(W^{\text{con}})$ (panel b). Rows and columns are ordered by cell type using the first-occurrence vocabulary of *load_zebrafish_hd_connectome*, with *IPNd** and *IPNds** contiguously at the top-left (the HD ring block), followed by *RIPN** and *pt-IPN** (afferent blocks). The matrix is sparse (density 0.027, 14 391 non-zero edges out of $731^2 = 534\,361$).

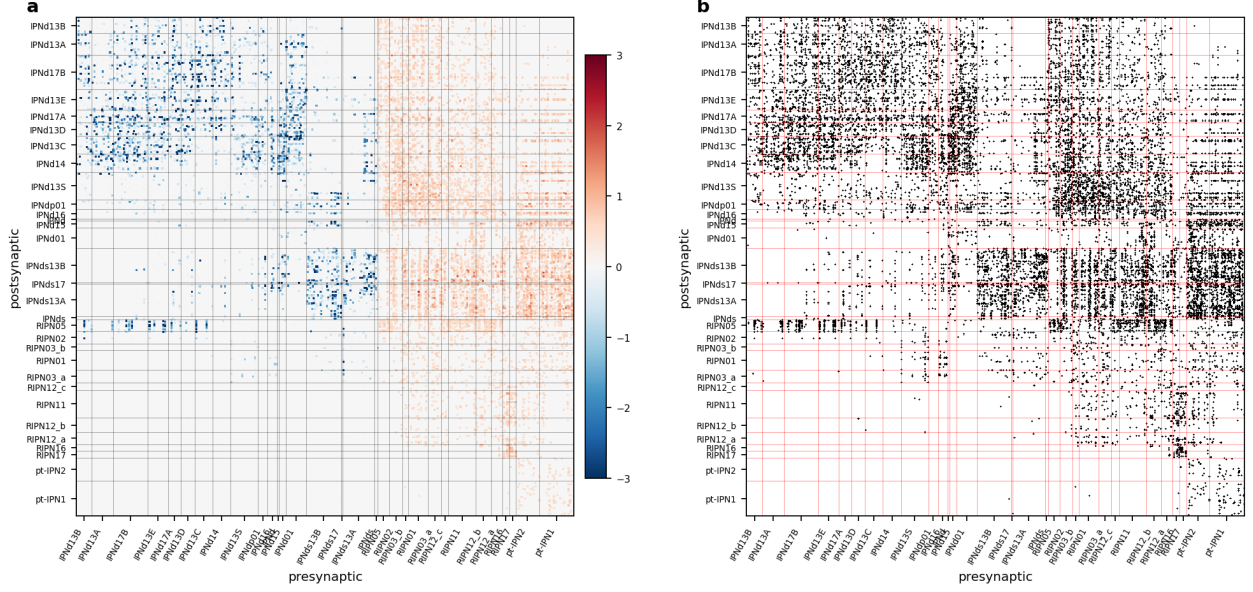


Figure 2: **Zebrafish HD-circuit connectome.** Matrix size $N \times N = 731 \times 731$, 14,391 non-zero edges (density = 0.027), $E : I = 7,663 : 6,728$ (53.2% / 46.8%). **(a)** Signed W^{con} , z -scored over non-zero entries and clipped to ± 3 ; entries are dilated by one pixel for visibility at panel resolution; neuron-type blocks marked on both axes; red = excitatory, blue = inhibitory. **(b)** Binary support $\text{supp}(W^{\text{con}})$ (black = edge present, white = absent, one-pixel dilation), block boundaries in red. The dorsal-IPN block (top-left) is the densest sub-matrix; the $RIPN^*$ and $pt\text{-}IPN^*$ rows show the afferent projection pattern onto the HD ring.

3 Methods

Recurrent neural network. The RNN model is a sign-locked discrete-time recurrent network whose connectivity is constrained to the larval-zebrafish HD-circuit connectome (Sect. 2). For each neuron $i \in \{1, \dots, N\}$ with $N = 731$ the subthreshold activity $\hat{h}_i(t)$ evolves according to

$$\tau \frac{d\hat{h}_i(t)}{dt} = -\hat{h}_i(t) + \sum_j \hat{W}_{ij}^{\text{rec}} \sigma(\hat{h}_j(t)) + \sum_l \hat{W}_{il}^{\text{in}} u_l(t) + \hat{b}_i, \quad (1)$$

$$\hat{y}(t) = (\cos \hat{\theta}(t), \sin \hat{\theta}(t)) = \hat{W}^{\text{out}} \sigma(\hat{h}(t)) + \hat{b}^{\text{out}}, \quad (2)$$

where σ is the sigmoid, $\tau = 0.1$ s, $\Delta t = 0.01$ s, and the input $u(t) \in \mathbb{R}^3 = (s(t), \cos \theta_0 \delta_{t,0}, \sin \theta_0 \delta_{t,0})$ carries a swim-triggered angular impulse $s(t)$ and a one-frame initial-heading cue. In Petrucco et al. 2023 Fig. 3c the network phase advances by $\approx \pi/4$ rad per swim ($Q_1, Q_3 = 0.49, 1.28$ rad, $n = 31$ fish), so $s(t)$ is a sparse stream of signed impulses of typical magnitude $\pi/4$ at swim onsets. Outside swim events $s(t) = 0$, mirroring the persistent silence of the $r1\pi$ bump in non-swimming fish (Petrucco et al. 2023 Fig. 3a, c). The readout from Eq. (2) reads from the HD-ring-only block of N neurons unless stated otherwise.

The recurrent matrix is sign-locked to the connectome template $W^{\text{con}} \in \mathbb{R}^{N \times N}$: the trainable per-edge parameter is the magnitude $\hat{S} \in \mathbb{R}^{N \times N}$, and

$$\hat{W}_{ij}^{\text{rec}} = |\hat{S}_{ij}| \text{sign}(W_{ij}^{\text{con}}) (1 - \delta_{ij}). \quad (3)$$

W^{con} comes out of *load_zebrafish_hd_connectome* with the Dale convention of Sect. 2: the outgoing columns of the GABAergic HD-ring cells ($IPNd^*$, $IPNds^*$) are negated and amplified by $5\times$, after which the matrix is rescaled to spectral radius 0.9. *All* sign-flipped columns belong to the bump population itself; there is no separate inhibitory cell type providing the long-range ring-shaping inhibition.

The swim-triggered drive is anatomically restricted to the afferent populations through a four-scalar gate on the first column of $\hat{W}^{\text{in}}$:

$$\hat{W}_{i,s}^{\text{in}} = v_q \quad \text{if } i \in q, \quad q \in \{\text{RIPN}_L, \text{RIPN}_R, \text{ptIPN}_L, \text{ptIPN}_R\}, \quad (4)$$

with four learnable scalars $\{v_{\text{RIPN}_L}, v_{\text{RIPN}_R}, v_{\text{ptIPN}_L}, v_{\text{ptIPN}_R}\}$ initialised antisymmetrically across hemispheres so that left swims and right swims drive the bump in opposite directions. The subpopulation membership comes from *afferent_subpop_ix* returned by the loader (sizes 102/101/41/44). The two remaining input columns — carrying the initial-heading impulse $(\cos \theta_0, \sin \theta_0) \delta_{t,0}$ — stay free for all neurons. RIPN^* is functionally implicated as the swim-triggered driver of the bump in Petrucco et al. 2023; pt-IPN^* is included in the gate so the optimiser can additionally route motor efference through the pretectal afferents if that improves heading recovery.

The learnable parameters are

$$\{\hat{S}, v_{\text{RIPN}_L}, v_{\text{RIPN}_R}, v_{\text{ptIPN}_L}, v_{\text{ptIPN}_R}, \hat{W}_{:, \theta_0}^{\text{in}}, \hat{b}, \hat{W}^{\text{out}}, \hat{b}^{\text{out}}\};$$

the connectome template W^{con} , the RIPN / pt-IPN subpopulation membership, the neuron-type assignments, τ and Δt are given. Training optimises a per-frame MSE on the heading readout with two connectome-prior regularisers and an ℓ_1 sparsity prior on the recurrent magnitudes,

$$\mathcal{L} = \frac{1}{BT} \sum_{b,t} \|\hat{y}_{b,t} - y_{b,t}\|_2^2 + \lambda_{\text{cos}} \mathcal{L}_{\text{cos}} + \lambda_{\text{nf}} \mathcal{L}_{\text{nf}} + \lambda_1 \|\hat{S}\|_1, \quad (5)$$

with block losses computed over type-pair masks $\mathcal{B}_p \subset \{1, \dots, N\}^2$ (one block per ordered pre/post neuron-type pair $p \in \mathcal{P}$),

$$\mathcal{L}_{\text{cos}} = \frac{1}{|\mathcal{P}|} \sum_{p \in \mathcal{P}} \left[1 - \frac{\langle \hat{W}_{\mathcal{B}_p}^{\text{rec}}, W_{\mathcal{B}_p}^{\text{con}} \rangle}{\|\hat{W}_{\mathcal{B}_p}^{\text{rec}}\| \|W_{\mathcal{B}_p}^{\text{con}}\|} \right], \quad (6)$$

$$\mathcal{L}_{\text{nf}} = \frac{1}{|\mathcal{P}|} \sum_{p \in \mathcal{P}} \text{ReLU}(\kappa - |\hat{W}_{\mathcal{B}_p}^{\text{rec}}|)^2. \quad (7)$$

The cosine-distance term $\mathcal{L}_{\text{cos}}$ anchors each block to the connectome template; the norm-floor term $\mathcal{L}_{\text{nf}}$ prevents any type-pair block from collapsing to zero by penalising the squared shortfall of the block mean absolute weight $|\hat{W}_{\mathcal{B}_p}^{\text{rec}}|$ below a fixed floor κ (typical value $\kappa = 0.05$). Because $\mathcal{L}_{\text{cos}}$ is scale-invariant, $\mathcal{L}_{\text{nf}}$ is required to prevent a degenerate solution in which the cosine prior is satisfied by an arbitrarily small recurrent magnitude. The type-pair vocabulary $\mathcal{P}$ is the cartesian product of the 31 fish2 cell types identified in Sect. 2, restricted to pairs with at least one non-zero entry in W^{con} . A linear-reconstruction analysis of the null space of the trained $\hat{W}^{\text{rec}}$ — a per-cell-type and per-unit sparsification test — is deferred to future work.

Graph neural network model. The GNN model replaces the dense recurrent operator $\sum_j \hat{W}_{ij}^{\text{rec}} \sigma(\hat{h}_j)$ of Eq. (1) with a message-passing graph neural network (Gilmer *et al.* 2017), and does *not* hard-code the leak or the firing-rate nonlinearity. The GNN imposes only the connectome topology of Sect. 2 and the input/output channels of Eqs. (1) and (2); the leak $-\hat{h}_i$ and the rate non-linearity σ must

be *discovered* inside the learned drift. Nodes correspond to neurons and edges to the non-zero entries of the zebrafish HD connectome template W^{con} . The GNN learns a latent embedding $\mathbf{a}_i \in \mathbb{R}^d$ per neuron, representing compact cell-type-specific identity; $d = 4$ is our default, validated once the training runs are in place. Subthreshold activity evolves as

$$\tau \frac{d\hat{h}_i(t)}{dt} = f_\theta\left(\hat{h}_i(t), \mathbf{a}_i, \sum_{j \in \mathcal{N}_i} \hat{W}_{ij} g_\phi(\hat{h}_j(t), \mathbf{a}_j)\right) + \sum_l \hat{W}_{il}^{\text{in}} u_l(t), \quad (8)$$

with the same readout (Eq. (2)), swim-gated encoder (Eq. (4)), and dynamics constants $(\tau, \Delta t)$ as the RNN. The neighbour set is fixed by the connectome topology, $\mathcal{N}_i = \{j : W_{ij}^{\text{con}} \neq 0\}$. The node update f_θ and edge message g_ϕ are three-layer perceptrons (width 64, ReLU, linear output). g_ϕ maps the presynaptic state $\hat{h}_j$ to a per-edge message conditioned on $\mathbf{a}_j$ and weighted by the per-edge gain $\hat{W}_{ij}$; f_θ then processes the postsynaptic state $\hat{h}_i$, the embedding $\mathbf{a}_i$, and the aggregated synaptic drive. We constrain g_ϕ to be monotonically increasing in $\hat{h}_j$ via the prior $\partial g_\phi / \partial \hat{h} \geq 0$ (see Eq. (10)); combined with the sign-locked $\hat{W}_{ij}$ (Eq. (9)) this restores Dale’s law at the per-edge level without introducing a $\pm$ -sign degeneracy. The leak, the firing-rate non-linearity, and any departure from a linear synapse must emerge from $(f_\theta, g_\phi, \mathbf{a})$ themselves; the monotonicity priors below provide the minimum inductive bias for f_θ, g_ϕ to behave like a neuron (postsynaptic drive increases with net excitatory input). The per-edge weight is sign-locked to the connectome as in the RNN,

$$\hat{W}_{ij} = |W_{ij}| \text{sign}(W_{ij}^{\text{con}}) (1 - \delta_{ij}), \quad (9)$$

where $W_{ij} \in \mathbb{R}$ is the trainable per-edge magnitude. In the zebrafish circuit $\text{sign}(W_{ij}^{\text{con}})$ is negative for every edge whose presynaptic neuron is in the bump ring ($IPNd^*$ or $IPNds^*$) and positive for every edge from $RIPN^*$ or $pt-IPN^*$; the GNN therefore inherits the inhibitory dIPN ring directly from the sign pattern.

The learnable parameters are

$$\{\theta, \phi, \mathbf{a}, W, v_{\text{RIPN}_L}, v_{\text{RIPN}_R}, v_{\text{ptIPN}_L}, v_{\text{ptIPN}_R}, \hat{W}_{:, \theta_0}^{\text{in}}, \hat{b}, \hat{W}^{\text{out}}, \hat{b}^{\text{out}}\};$$

the connectome template W^{con} , the RIPN / pt-IPN subpopulation membership, the neuron-type assignments, $\tau, \Delta t$, and the mediolateral ring ordering used by the decoder are given.

As with the RNN, heading recovery from cos/sin readouts is fundamentally non-unique: many distinct $(\theta, \phi, \mathbf{a}, W)$ configurations produce identical readouts. We constrain the solution space with the same connectome-prior regularisers as the RNN ($\mathcal{L}_{\text{cos}}, \mathcal{L}_{\text{nf}}$ from Eqs. (6) and (7), computed on the per-edge gain surface $\hat{W}_{ij}$ with floor threshold κ), together with sparsity and monotonicity priors on the MLPs:

$$\begin{aligned} \mathcal{L} = & \frac{1}{BT} \sum_{b,t} \|\hat{y}_{b,t} - y_{b,t}\|_2^2 + \lambda_{\text{cos}} \mathcal{L}_{\text{cos}} + \lambda_{\text{nf}} \mathcal{L}_{\text{nf}} \\ & + \lambda_0 \|\theta\|_1 + \lambda_1 \|\phi\|_1 + \lambda_2 \|\theta\| + \lambda_3 \|\phi\| + \gamma_0 \|W\|_1 + \gamma_1 \|W\| \\ & + \mu_0 \left\| \text{ReLU}\left(+\frac{\partial f_\theta(\hat{h}, \mathbf{a}, m)}{\partial \hat{h}}\right) \right\| + \mu_1 \left\| \text{ReLU}\left(-\frac{\partial f_\theta(\hat{h}, \mathbf{a}, m)}{\partial m}\right) \right\| \\ & + \mu_2 \left\| \text{ReLU}\left(-\frac{\partial g_\phi(\hat{h}, \mathbf{a})}{\partial \hat{h}}\right) \right\| + \mu_3 \|g_\phi(h_\star, \mathbf{a}) - h_\star\|, \end{aligned} \quad (10)$$

where $m = \sum_{j \in \mathcal{N}_i} \hat{W}_{ij} g_\phi(\hat{h}_j, \mathbf{a}_j)$ is the aggregated synaptic input. The ℓ_1 penalties λ_0, λ_1 on the MLP weights bias f_θ, g_ϕ toward sparser, simpler functions; the ℓ_2 penalties λ_2, λ_3 prevent weight explosion. The leak-monotonicity term μ_0 penalises *positive* slopes of f_θ in the postsynaptic state $\hat{h}$,

encoding the prior that the drift contains a leak ($\partial f_\theta / \partial \hat{h} \leq 0$); μ_1 penalises negative slopes of f_θ in the aggregated input m , encoding the prior that postsynaptic drive increases with net excitatory input. Together, μ_0 and μ_1 force the discovered drift to be decreasing in self-voltage and increasing in synaptic input — the two qualitative signatures of Eq. (1). The term μ_2 enforces the analogous monotonicity prior on g_ϕ in the presynaptic voltage. The normalisation term μ_3 anchors g_ϕ at a reference subthreshold value h_* , removing the multiplicative degeneracy between g_ϕ and $\hat{W}$ in Eq. (8). Hyperparameters $\lambda_{\{0,1,2,3\}}$, $\gamma_{\{0,1\}}$, $\mu_{\{0,1,2,3\}}$ and the MLP widths/depths are tuned with an agentic optimisation procedure, with $r_{\text{roll},1k}$ on a deterministic swim-impulse rollout as the primary score.

Mutual-information estimators. We use mutual information to quantify how strongly each unit’s firing rate $r_i(t) = \sigma(\hat{h}_i(t))$ tracks the heading $\theta(t)$ (or, where stated, the angular velocity $\omega(t)$ or the swim-onset trace $s(t)$) over a held-out rollout. For a single neuron we wrap θ to $(-\pi, \pi]$ via $\theta \leftarrow \arg(e^{i\theta})$, discretise the wrapped angle into $n_\theta = 32$ equal-width angular bins and r into $n_r = 20$ equal-width bins, then compute the plug-in estimator in bits,

$$\hat{I}(r_i; \theta) = \sum_{b,c} \hat{p}(b, c) \log_2 \frac{\hat{p}(b, c)}{\hat{p}(b) \hat{p}(c)}, \quad (11)$$

where $\hat{p}$ is the empirical joint over the (θ -bin, r -bin) histogram. The construction matches the drosophila MI estimator (matching the construction in the companion *Drosophila* CX paper, Methods, *Mutual-information estimators*) so that the per-neuron and per-cell-type quantities of Sects. 4.1 and 4.2 are comparable across the two species. For a group of K neurons we use the cross-validated lower bound

$$\hat{I}_{\text{LB}}(\mathbf{r}_{1:K}; \theta) = H(\theta) - \frac{1}{|\mathcal{D}_{\text{te}}|} \sum_{(t, \theta) \in \mathcal{D}_{\text{te}}} -\log_2 q_\psi(\theta_b \mid \mathbf{r}(t)), \quad (12)$$

where q_ψ is an L_2 -regularised multinomial logistic regression with input $\mathbf{r} \in \mathbb{R}^K$ and output the n_θ heading bins, trained on a stratified 5-fold split.

Image-domain ROI sampling and $\Delta F/F_0$. To make the model’s population dynamics comparable to two-photon calcium imaging of the larval-zebrafish dIPN, we sample activity *from the rendered frames* of the dorsal-view animation, not from the latent state $\hat{h}_i(t)$. Each frame is a 3000×4380 -pixel canvas in which each neuron’s skeleton is drawn with a per-segment alpha proportional to its firing rate; we place four axis-aligned ROI rectangles R_k ($k \in \{\text{dIPN}_L, \text{dIPN}_R, \text{dsIPN}_L, \text{dsIPN}_R\}$) over the dorsal-IPN neuropil (drawn on frame 0 of *figures/zebrafish/3D_voltage_const/frame_0000.png*) and integrate the green channel of the canvas RGBA buffer:

$$I_k(t) = \frac{1}{|R_k|} \sum_{(x,y) \in R_k} \text{img}_t[y, x, \text{green}]. \quad (13)$$

The four traces $I_k(t)$ are then converted to a fractional change with respect to a post-init baseline, mirroring the Seelig / Kim / Green calcium-imaging convention:

$$F_0[k] = \langle I_k(t) \rangle_{t \geq t_{\text{init}}} + \varepsilon, \quad (14)$$

$$\Delta F/F_0(k, t) = \frac{I_k(t) - F_0[k]}{F_0[k]}, \quad (15)$$

with $t_{\text{init}} = 0.5$ s (drops the network start-up transient that would otherwise dominate the baseline) and $\varepsilon = 10^{-6}$. The per-pair population-vector position overlaid on the kinographs in red is the

activity-weighted centre of mass across each L/R pair, $\text{pos}_p(t) = (I_{p,L} y_{p,L} + I_{p,R} y_{p,R}) / (I_{p,L} + I_{p,R})$, i.e. the same Seelig/Hulse PVA reduced to two phase bins per pair. Image-domain sampling inherits all the projection biases of the microscope (depth-of-field, projection, downsample) and so the kinographs are directly comparable to Petrucco et al. 2023 Fig. 2 without having to invert the calcium kernel.

4 Results

4.1 Functional partitioning of the HD circuit

We use the trained *zebrafish_hd_si_dipn* RNN (configuration Eqs. (1), (2), (4) and (5)) to ask, for each of the $N = 731$ neurons in the model, whether it *represents* the heading-direction code, *drives* or *updates* the bump, or does neither. The analysis combines three per-neuron scores computed on a 120 s closed-loop swim rollout (seed = 0, $\Delta t = 0.01$ s, burn-in 5 s):

1. the heading mutual information $\mathcal{I}_i = I(\hat{h}_i; \theta)$ in bits, estimated with the plug-in histogram of Eq. (11) (Sect. 3) at $n_\theta = 32$ heading bins and $n_r = 20$ activity bins,
2. the readout magnitude $\mathcal{W}_i = \|\hat{W}_{:,i}^{\text{out}}\|_2$, identically zero for $i \geq 443$ since the decoder of Eq. (2) slices the first $443 = n_{\text{dIPN}}$ columns,
3. the swim-triggered modulation $\mathcal{S}_i = \text{rms}_{t \in [-0.5\text{s}, +0.5\text{s}]} \langle \hat{h}_i(t_o + t) - \hat{h}_i(t_o - 0.5\text{s}) \rangle_o / \sigma_{\text{quiet}}(\hat{h}_i)$, the root-mean-square of the swim-onset-locked spike-triggered average over a ± 0.5 s window, normalised by the standard deviation of $\hat{h}_i$ in the swim-free intervals between onsets. The rollout is the same controlled probe that drives the kinograph of Sect. 4.2: one $\pi/8$ rad left swim every $\Delta t = 0.3$ s for 120 s, giving ~ 400 onsets, the leading edges of $|\omega(t)|$ exceeding 5% of its peak. Compared with the Poisson training distribution this gives more onsets per second and a regular heading sweep, removing the variance from random impulse fractions.

We then split each axis at its sample median and assign every neuron to one of four mutually exclusive classes via a decision tree:

$$\begin{aligned}
i \in \text{R} &\iff \mathcal{I}_i \geq \mathcal{I}^* \ \& \ \mathcal{W}_i \geq \mathcal{W}^* \\
i \in \text{L} &\iff \mathcal{I}_i \geq \mathcal{I}^* \ \& \ \mathcal{W}_i < \mathcal{W}^* \\
i \in \text{D} &\iff \mathcal{I}_i < \mathcal{I}^* \ \& \ \mathcal{S}_i \geq \mathcal{S}^* \\
i \in \text{Z} &\iff \mathcal{I}_i < \mathcal{I}^* \ \& \ \mathcal{S}_i < \mathcal{S}^*,
\end{aligned} \tag{16}$$

where the thresholds were $\mathcal{I}^* = 2.80$ bits, $\mathcal{W}^* = 0.066$ (median over the 443 dIPN columns only), and $\mathcal{S}^* = 0$. Class R is the *readout-relevant bump representation* (high HD content, large $\hat{W}^{\text{out}}$ column), class L the *latent / redundant* bump-correlated cells (high HD content but $\hat{W}^{\text{out}}$ effectively zero), class D the *drivers / updaters* (no HD content but strongly swim-modulated), and class Z is the *uninvolved / leaky* remainder.

Fig. 3 maps the four classes back to the 3-D anatomy and Tab. 2 summarises the per-cell-type breakdown. Under the controlled periodic-L probe stimulus the network ends up in classes R/L/D/Z = 156/210/365/0. The architectural priors of Sect. 3 hold as cleanly as before: *all* 51 *pt-IPN1* except two (latent), *all* 34 *pt-IPN2*, *all* 10/10 *RIPN17* and 10/10 *RIPN16* fall into the driver class D — exactly the afferent populations whose only access to the swim signal $s(t)$ is through the four-scalar PEN gate of Eq. (4). Conversely *no* *pt-IPN* or *RIPN* cell ends up in R (0/170), even though the slicing in Eq. (2) is purely positional and the optimiser is free to push readout mass onto

any dIPN cell. The bulk of R is carried by a small set of mediolateral dIPN subtypes: *IPNd13E* (24/30), *IPNd13C* (21/26), *IPNds13A* (18/47), *IPNd17B* (then 9/50), *IPNd13A* (17/32), *IPNds13B* (17/51) and *IPNd14* (16/27) account for $113/156 = 72\%$ of the R cells. This concentration — ~ 50 neurons supplying $> 50\%$ of $\|\hat{W}^{\text{out}}\|_1$, in agreement with the heavy-tailed column-norm distribution of $\hat{W}^{\text{out}}$ described in Sect. 3 — emerges *without* any explicit ℓ_1 prior on $\hat{W}^{\text{out}}$: the ring-attractor geometry plus the per-frame MSE on the cosine/sine readout makes most dIPN columns mutually redundant.

The *latent* class L is now 210 cells and is dominated by *IPNd17B* (20/50), *IPNd13S* (15/40), *IPNds13A/B* (27/98) and the surprise entry *RIPN05* ($13/17 = 76\%$ in L): unlike the rest of the habenular afferents this subtype acquires high HD mutual information through recurrent integration but is invisible to the linear readout because it sits outside the dIPN slice of Eq. (2). *RIPN11* (23/40) and *RIPN01* (14/29) also acquire substantial L membership, indicating that the regular probe makes more of the habenular afferents inherit a heading-correlated component via the recurrence than the Poisson training distribution did.

Under this regular probe the leak class Z *vanishes*: the periodic stimulus persistently drives the network and the recurrent connectivity propagates the drive to every cell with non-zero outgoing edges. There are no functionally idle units. The swim-modulation median therefore drops to $\mathcal{S}^* = 0$ and every below-median-MI cell crosses the $\mathcal{S}$ threshold by definition, so the Z partition is empty. The complementary D = 365 population now includes *IPNd01* (21/30, preserved), the entire *RIPN12_a* subtype (14/17), and a substantial fraction of the originally-leak *IPNd13B/D*, *IPNd17A*, *IPNdp01* cells, all consistent with the controlled stimulus propagating through the connectome.

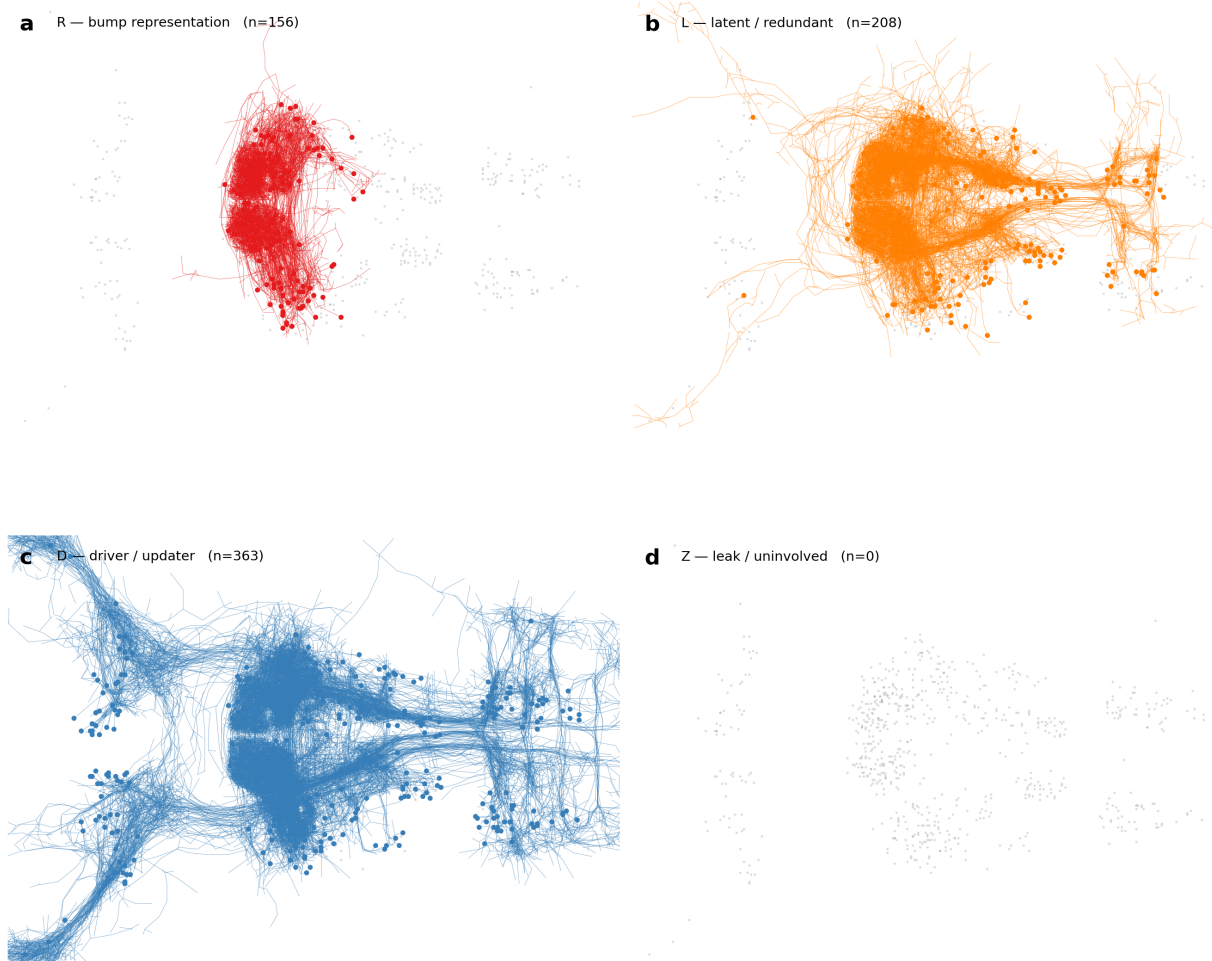


Figure 3: **Four functional classes of zebrafish HD-circuit neurons.** Dorsal view: each panel shows the skeletons and somata of the neurons assigned to one class via Eq. (16), with the somata of the other three classes overlaid as faint grey dots for spatial context. **(a)** R (red, $n = 156$): readout-relevant bump representation — exclusively dIPN, concentrated in the *IPNd13/14* and *IPNds13* sub-clusters. **(b)** L (orange, $n = 208$): latent bump-correlated cells that encode heading but are sliced out of the linear readout (notably *RIPN05* and *RIPN11*). **(c)** D (blue, $n = 363$): swim-locked drivers — all *pt-IPN* and most *RIPN* cells fall here. **(d)** Z (grey, $n = 0$): empty under the periodic probe stimulus — regular swim impulses propagate through the recurrent network and drive every cell. Thresholds: $\mathcal{I}^* = 2.80$ bits, $\mathcal{W}^* = 0.066$ (median over $\hat{W}^{\text{out}}$ columns), $\mathcal{S}^* = 0$. Per-cell counts and the model-index $\rightarrow$ bodyId mapping are written to *figures/zebrafish/fig_zebrafish_four_classes.csv*.

Table 2: **Per-cell-type breakdown across the four functional classes of Eq. (16)** under the periodic-L probe stimulus. Rows are sorted by total count; classes total R=156, L=210, D=365, Z=0 over $N = 731$ model neurons. Categories: I = *IPNd*, Is = *IPNds*, R = *RIPN*, P = *pt-IPN*.

cat	type	R	L	D	Z	total
Is	IPNds13B	17	13	21	0	51
P	pt-IPN1	0	2	49	0	51
I	IPNd17B	9	20	21	0	50
Is	IPNds13A	18	14	15	0	47
R	RIPN11	0	23	17	0	40
I	IPNd13S	9	15	16	0	40
P	pt-IPN2	0	0	34	0	34
I	IPNd13A	17	5	10	0	32
I	IPNd01	0	9	21	0	30
I	IPNd13E	24	1	5	0	30
R	RIPN01	0	14	15	0	29
I	IPNd14	16	2	9	0	27
I	IPNd13C	21	1	4	0	26
I	IPNd13B	9	7	7	0	23
R	RIPN12_b	0	1	21	0	22
I	IPNd13D	9	7	4	0	20
I	IPNd17A	5	5	9	0	19
R	RIPN03_a	0	12	7	0	19
R	RIPN02	0	9	10	0	19
R	RIPN05	0	13	4	0	17
R	RIPN12_a	0	3	14	0	17
I	IPNd01	1	6	7	0	14
I	IPNd16	0	9	5	0	14
R	RIPN12_c	0	5	6	0	11
R	RIPN17	0	0	10	0	10
R	RIPN16	0	0	10	0	10
I	IPNd15	0	7	3	0	10
R	RIPN03_b	0	6	3	0	9
Is	IPNds	1	0	4	0	5
I	IPNd	0	1	2	0	3
Is	IPNds17	0	0	2	0	2

4.2 Preferred-heading topography of the HD-coding cells

Among the 366 neurons in classes RUL identified above, we ask whether the network has self-organised a graded *preferred-heading* representation across the dIPN neuropil. We run the same periodic-L probe stimulus as in Sect. 4.1 ($\pi/8$ rad left swim every $\Delta t = 0.3$ s, 100 s, 4 s burn-in, $\theta_0 = 0$); the discrete heading jumps advance the bump in steady 22.5° steps and the ground-truth heading sweeps the full circle 20 times in the analysis window. For each neuron i we bin $\hat{h}_i(t)$ by $\theta(t)$ into $n_\theta = 36$ heading bins (10° wide) to form a tuning curve $\bar{r}_i(\theta_k)$, then compute the centred resultant

$$\rho_i e^{i\phi_i} = \sum_{k=1}^{n_\theta} (\bar{r}_i(\theta_k) - \langle \bar{r}_i \rangle) e^{i\theta_k}, \quad \mathcal{T}_i = \frac{|\rho_i|}{\sum_k |\bar{r}_i(\theta_k) - \langle \bar{r}_i \rangle|} \in [0, 1]. \quad (17)$$

ϕ_i is the cell's preferred heading and $\mathcal{T}_i$ its *specificity*: $\mathcal{T}_i = 1$ corresponds to a δ -shaped tuning curve, $\mathcal{T}_i = 0$ to a uniform or bimodal one. The R/L population is sharply unimodal — the empirical

distribution of $\mathcal{T}_i$ has a tight peak at 0.75 with a 10% shoulder below 0.71. We discard the shoulder by retaining only the $\mathcal{T}_i \geq \mathcal{T}^* = 0.70$ subset, leaving $350/366 = 96\%$ of $\mathbf{R} \cup \mathbf{L}$ as bona-fide HD-tuned cells with a unique preferred direction (up from 91% under the smoother constant-omega probe of the previous version of this figure; the discrete heading steps of the periodic probe match the network’s training regime more closely).

Fig. 4 maps the preferred angle ϕ_i of the 350 HD-tuned cells back to the 3-D anatomy. We partition $[-180^\circ, +180^\circ]$ into nine 40° wedges and render one panel per wedge, drawing the somata and arbors of cells whose ϕ_i falls in that wedge in the wedge’s central hue (cyclic HSV) on a dark-grey backbone of every available SWC. Each wedge contains $n = 21$ to 67 cells (mean = 39), with the wedges that overlap $\phi \approx \pm 180^\circ$ and 0° slightly over-represented because the left swims drive the bump through those phases first.

The figure makes one substantive negative observation explicit: the preferred angle is *not* laid out as a spatial coordinate over the dIPN neuropil. Soma dots for every 40° wedge fill essentially the same dIPN region, and the active patch does not translate as ϕ sweeps from panel (a) to panel (i). This is quantitative: regressing soma mediolateral position x_i on the preferred phase within each hemisphere gives $\text{Pearson}(x_i, \cos \phi_i) = -0.01$ (L), -0.02 (R) and $\text{Pearson}(x_i, \sin \phi_i) = +0.02$ (L), -0.10 (R); the circular-circular mean resultant of (soma angle about the hemisphere centroid – preferred angle) is 0.25 (L) and 0.29 (R), close to the value expected for random preferred-angle assignment. The HD-tuned cells therefore form a sharply unimodal code in firing-rate space (specificity $\mathcal{T}_i \approx 0.75$, Eq. (17)) but their preferred directions are *interspersed* across the neuropil rather than topographically ordered, in contrast to the fly EPG ring on the ellipsoid body where preferred heading is one-to-one with anatomical position. The ring-attractor signature that the trained model does carry lives in *activity space*: PC1–PC2 of the IPNd+IPNd population traces a closed circle as the heading sweeps (companion analysis `fig_zebrafish_pca_ring.py`). This is consistent with Petrucco et al. 2023 Fig. 1e–f, where the dIPN ring is reported in functional similarity-space rather than as an anatomical topography across cell-body positions.

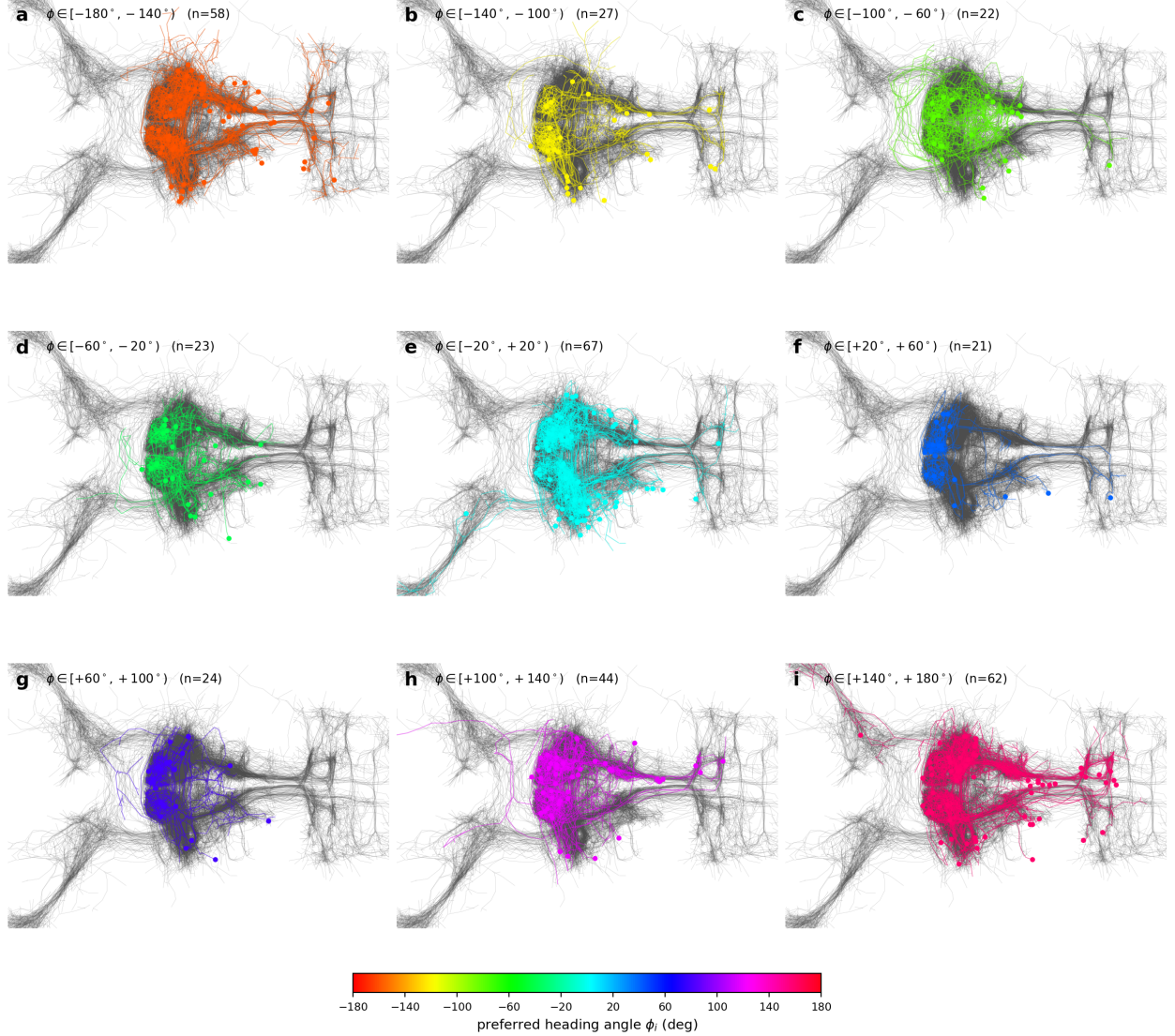


Figure 4: **Preferred-heading code is sharp in firing rate but not topographic in anatomy.** Periodic-L probe rollout (one $\pi/8$ rad left swim every $\Delta t = 0.3$ s for 100 s) is used to estimate each neuron's tuning curve and the centred resultant of Eq. (17). The 350 cells in $R \cup L$ with specificity $\mathcal{T}_i \geq 0.70$ are partitioned into nine 40° -wide preferred-angle wedges and rendered one wedge per panel: somata (dots) and skeleton arbors of cells in that wedge, drawn in the wedge's central hue (cyclic HSV), on top of a dark-grey backbone of every available SWC. Panel letters (a–i) sweep from $\phi \in [-180^\circ, -140^\circ)$ through $[-20^\circ, +20^\circ)$ to $[+140^\circ, +180^\circ)$; the cell count per wedge is shown next to each panel letter. The soma dots for every wedge fall in the same dIPN region: preferred heading is not laid out as a spatial coordinate over the neuropil (Pearson($\text{soma}_x, \cos \phi$), $\sin \phi$) correlations ≤ 0.10 within each hemisphere). The shared bottom colour bar shows the cyclic phase scale and the bin edges. Per-cell values (ϕ_i and $\mathcal{T}_i$, with the kept/discarded flag) are written to *figures/zebrafish/fig_zebrafish_pref_angle.csv*.

4.3 Tuning-curve sharpness of the HD-coding cells

Fig. 4 shows where each HD-coding cell sits on the preferred-heading circle but says nothing about how *sharp* that preference is. Fig. 5 answers the sharpness question by overlaying the centred tuning curves of every HD-coding cell and comparing the population profile to a parametric reference bump.

Factual results. Of the $n = 366$ cells in the $R \cup L$ classes, **350 (96%) pass the specificity threshold $\mathcal{T}^* = 0.70$** ; the histogram of $\mathcal{T}$ (Fig. 5a) peaks tightly around 0.75 and has essentially no mass below 0.5. The 350 retained cells, once centred on their preferred angle and normalised to unit peak, collapse onto a single sharp bump (Fig. 5b): the population mean has full width at half maximum (FWHM) of $\approx 100^\circ$ and a tight 25–75 percentile band. A unit-peak von Mises with concentration $\kappa \approx 2$ tracks the mean within the band over the full $\pm 180^\circ$ range. Together, panels a and b say that the trained network’s HD code is not just present (panel b) but well-defined and near-sinusoidal (panel c) at the per-cell level. Panel a shows a single representative tuning curve for context.

Computation, panel a (single-neuron tuning curves). Panel a overlays the tuning curves $\bar{r}_i(\theta_k) - \langle \bar{r}_i \rangle$ of three example neurons. The per-cell mean is subtracted (but no peak-normalisation is applied) so all three curves share a zero baseline and their relative amplitudes are preserved; without this subtraction the different DC offsets between cells dominate the y -axis and the actual bumps are invisible. The two specific cells (blue and green) sit near the median specificity of the kept population ($\mathcal{T} \approx 0.75, 0.73$) but have preferred angles roughly 180° apart, so the reader can see that a high $\mathcal{T}$ does not commit a cell to one particular ϕ . The non-specific cell (grey, $\mathcal{T} = 0.18$) shows what $\mathcal{T} \ll \mathcal{T}^*$ looks like in rate space: an essentially flat curve with no resolvable peak. Dashed vertical lines mark each cell’s preferred angle ϕ_i , computed as the argument of the circular resultant of the centred curve (same formula as the specificity numerator below). This panel is the single-cell view of the population summaries in panels b–d: the specific cells live in the right peak of the histogram in panel b, contribute one curve each to the overlay in panel c, and place one count each in the histogram of panel d; the non-specific cell sits in the left tail of panel b and is excluded from panels c–d.

Computation, panel b (specificity histogram). For every cell i we compute a tuning curve $\bar{r}_i(\theta_k)$ by binning the rollout heading $\theta(t)$ into $K = 36$ equal bins of 10° and averaging the cell’s activity $h_i(t)$ over the time points whose heading falls in each bin (*fig_zebrafish_pref_angle._tuning_curves*). The *specificity*

$$\mathcal{T}_i = \frac{|\sum_k (\bar{r}_i(\theta_k) - \langle \bar{r}_i \rangle) e^{i\theta_k}|}{\sum_k |\bar{r}_i(\theta_k) - \langle \bar{r}_i \rangle|} \in [0, 1]$$

is the ratio of the magnitude of the resultant (first Fourier coefficient) of the centred tuning curve to its ℓ_1 size. A flat curve gives $\mathcal{T} = 0$; a delta at one bin gives $\mathcal{T} = 1$; a pure cosine sits near $\mathcal{T} = 2/\pi \approx 0.64$. The histogram in panel a is over the $n = 366$ cells in the $R \cup L$ classes (HD-coding R and L from the four-classes partition, Fig. 3); the red dashed line at $\mathcal{T}^* = 0.70$ is the cutoff used in panel c and in all downstream HD-population analyses.

Computation, panel c (centred tuning curves). For each of the 350 cells that pass $\mathcal{T} \geq \mathcal{T}^*$: mean-subtract $\bar{r}_i$, circularly roll the curve so the peak bin sits at $\theta = 0^\circ$, and divide by the peak so the curve has unit height. All 350 normalised curves are drawn as thin grey lines. The bold red curve is their across-cell mean; the shaded red band is the across-cell 25–75 percentile envelope at

each angular bin. The FWHM annotation at the panel title is read off the mean curve by linear interpolation between bins crossing the 0.5 level.

Computation, panel d (preferred-angle distribution). Panel d counts how many HD-coding cells prefer each angular bin. For each cell that passes $\mathcal{T} \geq \mathcal{T}^*$ we take its preferred angle $\phi_i = \arg(\sum_k (\bar{r}_i(\theta_k) - \langle \bar{r}_i \rangle) e^{i\theta_k})$ — the same circular mean used in the specificity numerator, returning a phase in $(-\pi, \pi]$ — and bin ϕ_i in degrees into the same $K = 36$ equal bins of 10° used by the tuning curves. The red dashed horizontal line marks the uniform-coverage expectation $n_{\text{kept}}/K = 350/36 \approx 9.7$ cells per bin. The empirical distribution covers the full circle (no bin is empty: min = 3 cells, max = 28 cells), but it is not uniform: a chi-squared deviation from uniformity gives $\chi^2 \approx 154$ (df = 35, $p \ll 10^{-3}$), with two clear over-represented bands. This is the rate-space counterpart of Fig. 4: the ring is well-tiled (every 10° slice of heading has at least three cells that prefer it) but the tiling is uneven, with some preferred angles carried by $\sim 3\times$ as many cells as others. The over-represented bands near 0° and 180° are expected: the readout channels are $(\cos \theta, \sin \theta)$, so neurons that contribute most to the cos channel must tile the cos axis ($0^\circ/180^\circ$); the training initial heading θ_0 is itself uniform on $[0, 2\pi)$, so the symmetry breaking comes from the readout convention rather than the data distribution.

Why a von Mises fit and not a "regular" (Gaussian) fit. Heading direction lives on a circle: θ and $\theta + 360^\circ$ are the same heading, and any tuning curve must be a periodic function of θ with the same value at $\pm 180^\circ$. A Gaussian $\exp(-(\theta - \mu)^2/(2\sigma^2))$ ignores this: it sits on the real line, has different values at -180° and $+180^\circ$ unless artificially symmetrised, and its tails do not wrap around. The natural "Gaussian on a circle" is the *von Mises distribution*,

$$f(\theta; \mu, \kappa) \propto \exp(\kappa \cos(\theta - \mu)),$$

which is periodic by construction. In our figure the reference curve is normalised to unit peak, $f_\star(\theta; \kappa) = \exp(\kappa(\cos \theta - 1))$, so it shares the 0–1 scale of the centred normalised data (panel c's y axis). The two ingredients are (i) a preferred angle μ (centred to 0 before plotting) and (ii) a concentration κ : small κ gives a broad bump, large κ a sharp spike. We grid-search $\kappa \in [0.2, 10]$ for the value that minimises the ℓ_2 distance against the centred mean curve; the winner ($\kappa \approx 2$) is shown as the dashed black overlay. This fit is purely a visual reference — the empirical mean is the data — and it is included so the reader can tell at a glance that the population bump is well-approximated by the standard circular analogue of a Gaussian, and how broad it is on the natural circular scale. For large κ the von Mises reduces to a Gaussian via the small-angle expansion $\kappa(\cos \theta - 1) \approx -\kappa \theta^2/2$, with $\sigma^2 = 1/\kappa$; for $\kappa \approx 2$ we are in the explicitly circular regime where the Gaussian approximation would noticeably mis-fit the wraparound.

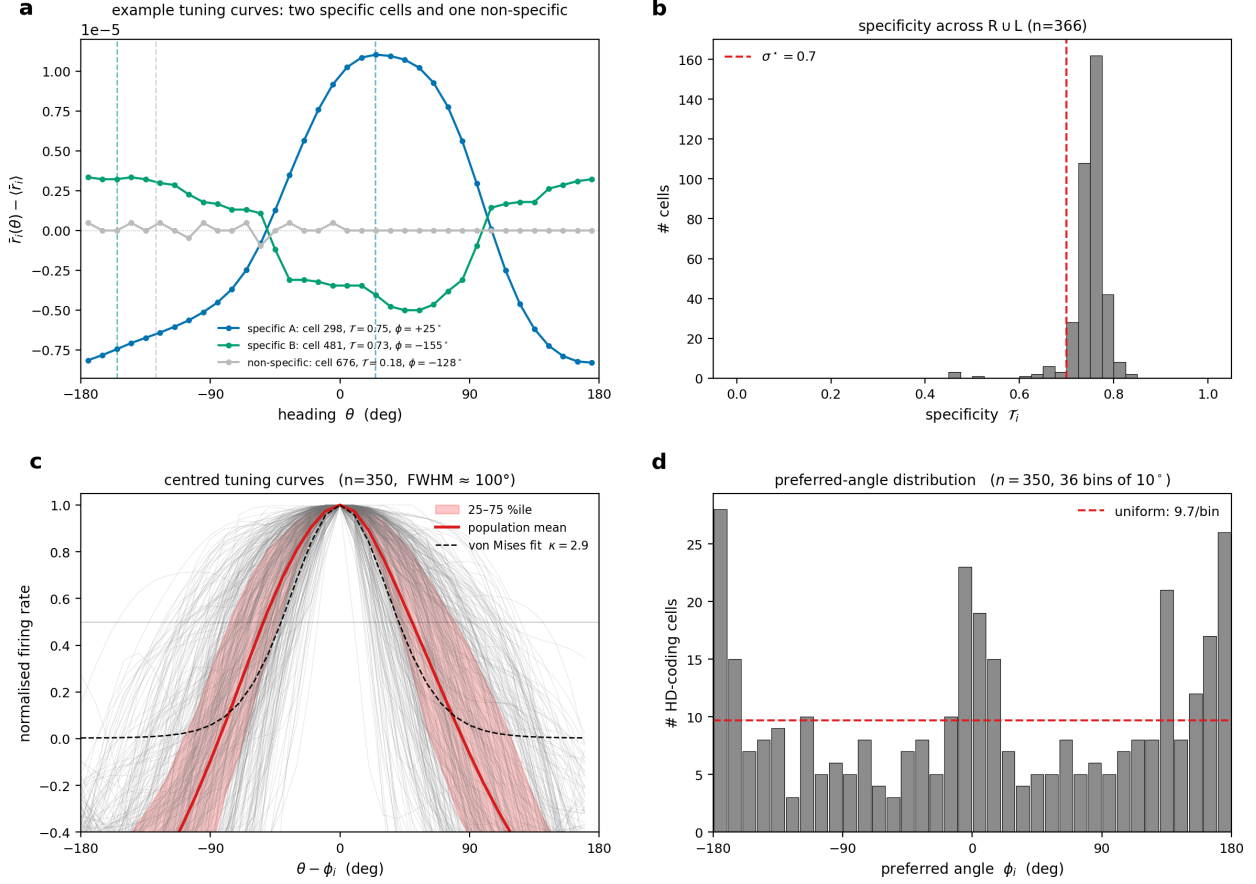


Figure 5: **Tuning-curve sharpness of HD-coding cells.** (a) Mean-subtracted tuning curves $\bar{r}_i(\theta) - \langle \bar{r}_i \rangle$ for three example cells: two specific cells (blue $\mathcal{T} = 0.75$, $\phi \approx +25^\circ$; green $\mathcal{T} = 0.73$, $\phi \approx -155^\circ$) and one non-specific cell (grey $\mathcal{T} = 0.18$, flat). Subtracting the per-cell mean puts all curves on a shared zero baseline while preserving relative amplitudes; dashed verticals mark each cell’s preferred angle in the matching colour. Panels b–d are population summaries derived from many such curves (the specific cells contribute, the non-specific cell is excluded by $\mathcal{T} \geq \mathcal{T}^*$). (b) Distribution of the specificity $\mathcal{T}_i$ over the $n = 366$ cells in $R \cup L$; dashed red line at $\mathcal{T}^* = 0.70$ marks the cutoff used downstream. The distribution clusters tightly around 0.75 with negligible mass below 0.5: most $R \cup L$ cells have a well-defined preferred heading. (c) Centred (peak at 0°) and unit-peak-normalised tuning curves of the 350/366 cells that pass the specificity cut, drawn as thin grey lines. Bold red curve: across-cell mean. Red band: across-cell 25–75 percentile envelope at each angular bin. Black dashed: ℓ_2 -fit unit-peak von Mises $f_*(\theta; \kappa) = \exp(\kappa(\cos \theta - 1))$ with concentration $\kappa \approx 2$, shown as a parametric reference (the empirical mean is the actual data). The population bump has FWHM $\approx 100^\circ$ on the circle. (d) Distribution of preferred angles ϕ_i across the 350 HD-coding cells that pass $\mathcal{T} \geq \mathcal{T}^*$, binned into the same $K = 36$ equal 10° bins. The red dashed line at ≈ 9.7 marks the uniform-coverage expectation n_{kept}/K . Every bin contains at least three cells (min = 3, max = 28, mean ≈ 9.7), so the full circle is tiled, but the tiling is significantly non-uniform ($\chi^2 \approx 154$, df = 35).

References

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